

The Critical Interface

ENGINEERING CASEBOOK ISSUE 004 · THE CRITICAL INTERFACE

17 AUG 2026 · 3-PAGE EDITION

DEEP DIVE · CASE 013

The half-inch plate
beneath a thousand feet
of bridge

I-35W Mississippi River Bridge · Minneapolis, USA · 2007
STEEL TRUSS GUSSET PLATES LOAD RATING CONSTRUCTION LOADING NTSB
HAR-08/03, Fig. 15

At 6:05 p.m. on 1 August 2007, about 1,000 ft of the I-35W
deck truss collapsed. Thirteen people died and 145 were
injured. The National Transportation Safety Board traced U10
initiation to the U10 gussets, connection plates designed at
roughly half the thickness required at several critical
nodes.[S1]

compression diagonal
pushes laterally as
bowed plate loses stability

YOU ARE THE ENGINEER

A concrete overlay is underway. About 578,735 lb (2.57 MN)
is staged in a compact footprint near U10. A global truss
check says the bridge can carry the work, but the model
Simplified truss
elevation and U10 connection zone
omits gusset capacity. Do you check total construction load,
or where it enters and which connection must transfer it?

The gusset was not a passive patch between strong members. It had to
collect
large, non-collinear forces and keep a compression diagonal geometrically
THE LOAD HISTORY stable.

1967 Bridge opens. Main-truss gussets are not independently load-

BOWING WAS THE WARNING AND THE IMPERFECTION

rated.
The unsupported U10 edge was about 30 in long. With a 0.5
1977 A thicker deck adds more than 3.0 million lb, raising dead load
by 13.4%. in plate, the edge slenderness was:

New barriers and railings add about 1.13 million lb, another

1998 $L / t = 30 / 0.5 = 60 > 48$

6.1%.
Photographs show bowing along unsupported edges of seven The AASHTO rule

required edge stiffening. Pre-collapse
1999/03

of eight U10/U10' plates. bowing measured 0.44 to 0.99 in, about the
plate thickness.

2007 Overlay materials are concentrated over the inner west side An
inspector noticed it but did not record it because it was
near U10. assumed to be a construction artifact. NTSB found no
The construction materials did not simply make the bridge "too analysis
of its cause or capacity effect.[S1]

heavy". NTSB found that the same load spread over the planned

INVESTIGATOR FINDING ENGINEERING INTERPRETATION

530 ft pour would not have overloaded the plates. The decisive
Sandia reproduced the observed lateral shift and found that
feature was load concentration, not gross tonnage.[S1]
initial bowing materially reduced the deficient joint's
planned pour 530 ft capacity. A flat 0.5 in model remained stronger than

the
actual staging near U10 collapse loading; a 1.0 in plate had ample
capacity. Bowing

The same gross material can be acceptable when distributed and critical
when was not awhenrT* ET BT /F1 6.1 Tf 7.32 TL ET BT 1 0 0 1 34.01575 589.56511.1rl0 ET sho aa6T

The Critical Interface

ENGINEERING CASEBOOK PROBLEMS FROM PRACTICE · ISSUE 004
SITE PROBLEM · CASE 014

014
Strong basalt, weak slope
Mumbai-Pune Expressway · Adoshi, India
The cut at km 41 exposed a roughly 40 m vertical basalt face.
Intact amygdaloidal and compact basalt tested around 150 and 180 MPa UCS, yet rockfall repeatedly reached the expressway during daylighting wedge
the monsoon. The contradiction disappears once the slope is treated as a rock mass, not a laboratory cylinder.[S1]
YOU ARE THE ENGINEER
Three joint sets intersect. Some planes daylight toward the road; a weathered subhorizontal layer erodes in water; an oblique expressway joint forms wedges; and a seasonal stream follows fractures
High intact strength does not prevent kinematic release along persistent, water-
above the cut. Which governs: intact UCS, or discontinuity pressurised discontinuities.
orientation, persistence, infill and water?
THE TOOLBOX
Loose blocks were scaled. The recommended treatment
Stereographic projection and the Markland test identified planar combined 70 mm steel-fibre shotcrete with a 300 x 300 mm and wedge kinematics. Blasting and incomplete dressing had cable net, anchored in a crest trench and fixed to the face exposed loose blocks, while intense rain entered open joints, with 0.5 m expansion or grouted bolts. Some sections used eroded the altered layer and undercut overhangs. Blocks were separating along surfaces that gravity already liked. net alone.[S1]
The system was not expected to stop every block. It controlled
ENGINEER'S NOTEBOOK - ROCK SLOPE TRIANGLE
trajectories and prevented bouncing or carriageway blockage.
Before strength: map joint sets, daylighting planes, wedge That is the difference between slope stabilisation and intersections, altered seams and crest drainage. A dry-season rockfall management.
visit can conceal the governing hydraulic load case.
SOURCES FOR THIS CASE
ENGINEERING INTERPRETATION Markland analysis is a kinematic screen, not a
S1 Kumar et al., Rockfall and Subsidence on Mumbai-Pune factor of safety. It shows whether a plane or wedge can move Expressway,
IJGCH 2(1), 2010.
geometrically; persistence, roughness, infill, water pressure, block volume
and support demand still determine whether it will.

DETAIL / MATERIAL · CASE 015
015
Stainless steel failed where pool water never touched it
Indoor swimming-pool roof collapse · Uster, Switzerland · 1985
Twelve people were killed when a concrete pool roof supported THE
MATERIAL-SELECTION TRAP
by stainless-steel tension rods collapsed. UK HSE guidance HSE warns that grades 1.4301 (304) and 1.4401 (316) can be identifies stress corrosion cracking (SCC) as the failure susceptible in these unwashed atmospheric conditions near 30 mechanism.[S1] C. Corrosion literature attributes the Uster hangers to The exposure was above the pool. Warm humid air carried molybdenum-free Type 304.[S2] The lesson is not "stainless is chlorides into cooler, poorly washed ceiling spaces. unsafe". It is that stainless is not an exposure class.
Condensation and drying concentrated salts on highly stressed, THE
TOOLBOX
inaccessible components. A grade adequate in pool water can Exact proprietary product: not reported. Select a grade therefore be unsuitable for a hidden hanger in the pool qualified for the atmosphere; reduce tensile and residual stress; atmosphere. avoid inaccessible crevices; control ventilation and chemistry;
and make critical hangers visible and replaceable.
ENGINEER'S NOTEBOOK - HIDDEN STAINLESS INVESTORY
susceptible tensile chloride
Record grade, function, stress state, access and cleaning
grade stress environment regime for every critical stainless hanger, bracket and fastener. Early SCC can be hard to detect; inspection frequency and material selection require competent corrosion
All three are required for SCC advice.
Applied load and residual stress from forming or welding can both supply the **SOURCES FOR THIS CASE**
tensile-stress leg of the SCC triangle. S1 UK HSE, SCC of Stainless S2 IMOA, Molybdenum Brings
PRACTICAL WARNING SCC can advance with little general corrosion. A Steels in Swimming Pool Safety into Swimming Pool component may retain a deceptively clean surface while the load-Buildings, SIM 5/2002/18 Buildings.
bearing section is badly cracked. Access and replaceability therefore (2002).
belong in the original detail, not only the maintenance manual.[S1]
2/3

The Critical Interface

ENGINEERING CASEBOOK ENGINEERING DONE WELL · ISSUE 004
ENGINEERING WIN · CASE 016
016